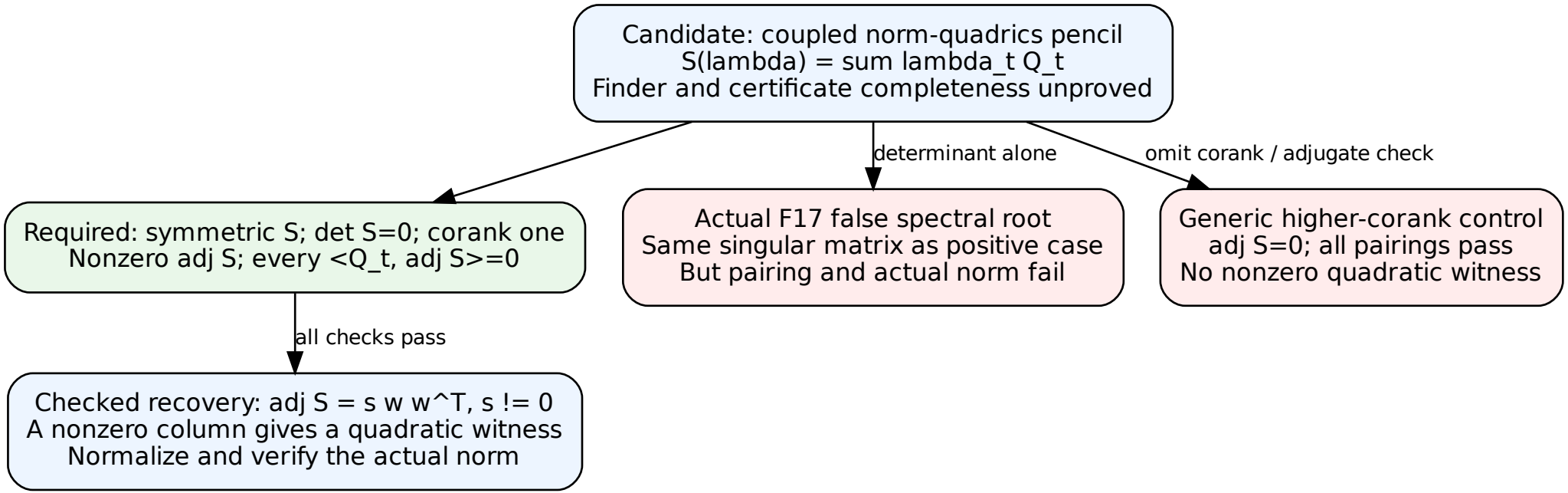
Candidate: coupled norm-quadrics pencil
S(lambda) = sum lambda_t Q_t
Finder and certificate completeness unproved
determinant alone
omit corank / adjugate check
Required: symmetric S; det S=0; corank one
Nonzero adj S; every <Q_t, adj S>=0
Actual F17 false spectral root
Same singular matrix as positive case
But pairing and actual norm fail
Generic higher-corank control
adj S=0; all pairings pass
No nonzero quadratic witness
all checks pass
Checked recovery: adj S = s w w^T, s != 0
A nonzero column gives a quadratic witness
Normalize and verify the actual norm